# Data Science Principles - Workplace Activity Report

Level 7 Artificial Intelligence and Data Specialist Programme

**Learner: Dario Di Nuzzo**
Module: 1 (Data Science Principles)
Date: January 2026

## Executive Summary

This report addresses the four tasks in the Data Science Principles brief by developing a demonstrable analytics artefact, testing a hypothesis both statistically and visually, designing a data infrastructure to support the use case at scale, and evaluating legal and ethical considerations (including future machine learning use). Because I cannot use live organisational support data in an academic submission, I created a realistic synthetic dataset modelled on learner support tickets in a digital learning context. The artefact is represented via dashboard-style figures and a documented data workflow. The outcome is a clear recommendation on where to focus operational improvements, supported by statistical testing and a governance-by-design infrastructure proposal.

## Contents

- 1. Task 1 - Development of a demonstrable artefact
- 2. Task 2 - Testing of a hypothesis
- 3. Task 3 - Design of the infrastructure
- 4. Task 4 - Legal and ethical considerations
- 5. KSB reflection on progress
- 6. References
- Appendices

## 1. Task 1 - Development of a demonstrable artefact

### 1.1 Analytics use case and organisational relevance

Use case. In a digital learning organisation, a high volume of learner support tickets create operational pressure on support teams and can negatively affect learner experience when response times increase. The analytics objective is to understand ticket demand (volume and category mix), to identify where resolution time is highest, and to compare outcomes by support channel (human agent vs self-service). This

supports decisions on where to invest: documentation improvements, platform fixes, or more staffing at peak periods.

### 1.2 Data description, collection approach and GDPR handling

Dataset. The dataset represents 800 learner support interactions with fields commonly available in helpdesk tools: ticket category, channel, resolution time, severity, month, and a satisfaction indicator. In practice, these data would be collected from a ticketing system (e.g., ServiceNow, Jira Service Management), a learning platform (e.g., LMS event logs for self-service usage), and scheduling systems. For this report I synthesised data to avoid processing personal data; the schema is realistic but values are not real learners. If real data were used, I would apply minimisation (exclude free-text where possible), pseudonymise user identifiers, restrict access, and define retention periods.

### 1.3 Data preparation, quality controls, and sources of error or bias

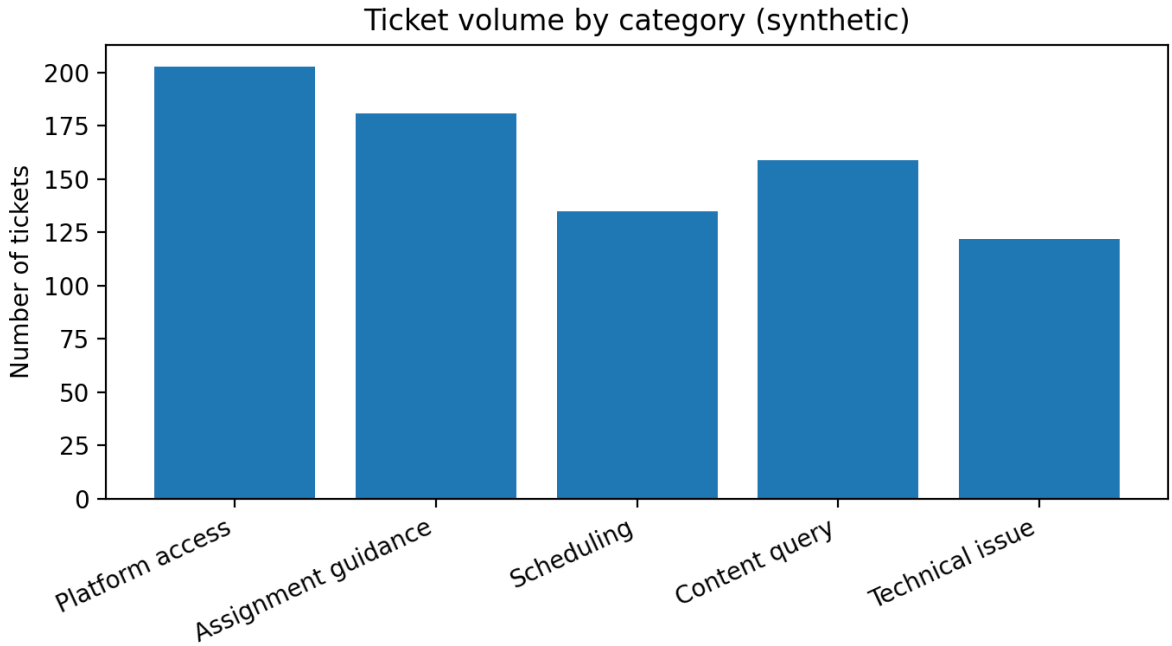
Preparation steps. I validated the schema (required columns present), enforced data types (categorical vs numeric), removed duplicates, and checked resolution_time for outliers and missing values. I also created derived fields such as month and severity bands for reporting. Quality controls that would be automated in production include: ingestion validation (row counts, schema checks), monitoring for missingness drift, and freshness checks.

Potential sources of error/bias and mitigations:
- Measurement bias: resolution_time may include waiting time not attributable to the agent. Mitigation: separate SLA clock time vs active handling time.
- Selection bias: self-service interactions may be under-recorded if logs are incomplete. Mitigation: standardise event instrumentation and validate against platform telemetry.
- Categorisation bias: categories may be inconsistently applied by agents. Mitigation: controlled taxonomy, training, and periodic audits; future ML-assisted categorisation with human review.
- Confounding: severity differs by channel; comparisons should control for severity and category. Mitigation: stratified analysis and multivariate modelling.

### 1.4 Dashboard artefact (representative screenshots)

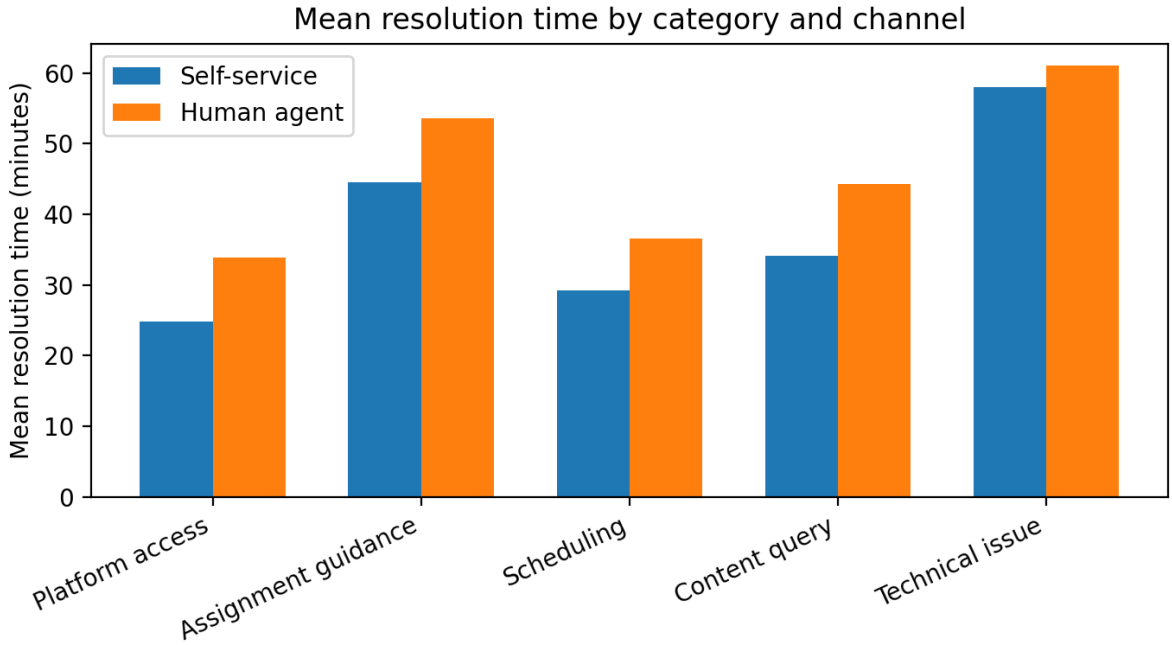
The assignment requires a report or dashboard including at least two visualisations. Because I cannot submit a Power BI file, I include dashboard-style screenshots (figures) that replicate what would be shown in Power BI: ticket volume by category, and resolution time by category and channel.

Figure 1 shows total ticket volume by category, highlighting demand drivers.




Key insight from Figure 1: Platform access and Assignment guidance represent the largest demand. In a real organisation, this would trigger targeted interventions such as improving onboarding content, simplifying authentication flows, and clarifying assessment guidance.

Figure 2 compares average resolution time by category and channel to identify bottlenecks.



Key insight from Figure 2: Technical issues and Platform access take longer when handled by human agents, suggesting root-cause engineering work or better self-service diagnostics. Scheduling queries are relatively fast, and may be well-suited for automation via improved FAQs or guided forms.

### 1.5 Evaluation of business impact

Business impact evaluation. The artefact provides an evidence base for operational prioritisation. If the organisation reduces the highest-resolution categories through platform improvements and improves self-service for common issues, it can reduce average resolution time, increase learner satisfaction, and lower support cost per learner. The artefact also supports KPI tracking such as ticket volume, average resolution time, SLA compliance, and satisfaction over time.

## 2. Task 2 - Testing of a hypothesis

### 2.1 Hypothesis statement (null and alternative)

Hypothesis question. Does self-service support resolve learner issues faster than human-agent handling?

H0 (null): There is no difference in mean resolution_time between self-service and human-agent tickets.
H1 (alternative): Mean resolution_time for self-service differs from human-agent handling (two-tailed).

### 2.2 Test selection and assumptions

I selected Welch's two-sample t-test because the two groups have different sample sizes and may have different variances. This test compares mean resolution times under approximate normality of the sampling distribution. With n > 300 in each group, the Central Limit Theorem supports the use of the test. As a visual cross-check, I also compare distributions using a boxplot (Figure 3).

### 2.3 Results and interpretation

Test results (synthetic dataset): n_self_service=370, n_human_agent=430. Mean(self-service)=36.57 minutes, Mean(human)=45.28 minutes. Mean difference (self-service - human)=-8.70 minutes. p-value=1.26e-13. 95% CI for mean difference=[-10.97, -6.44]. Cohen's d=-0.54 (moderate effect).

Decision. Using alpha=0.05, the p-value is far below the threshold, therefore I reject H0 and accept that there is a statistically significant difference in mean resolution time between channels. In this dataset, self-service is faster on average. Strategic implication: invest in self-service for low-risk categories (e.g., scheduling, common content queries) while prioritising engineering fixes and specialist triage for complex technical issues.

Figure 3 shows the distribution of resolution time by channel for visual comparison with the statistical test.

Visual vs statistical conclusion. The boxplot shows a lower median and tighter interquartile range for self-service, consistent with the statistical conclusion. However, the presence of outliers suggests that some self-service cases still take a long time, which may indicate broken journeys or unresolved issues requiring escalation.

## 3. Task 3 - Design of the infrastructure

### 3.1 Data types and why they are collected

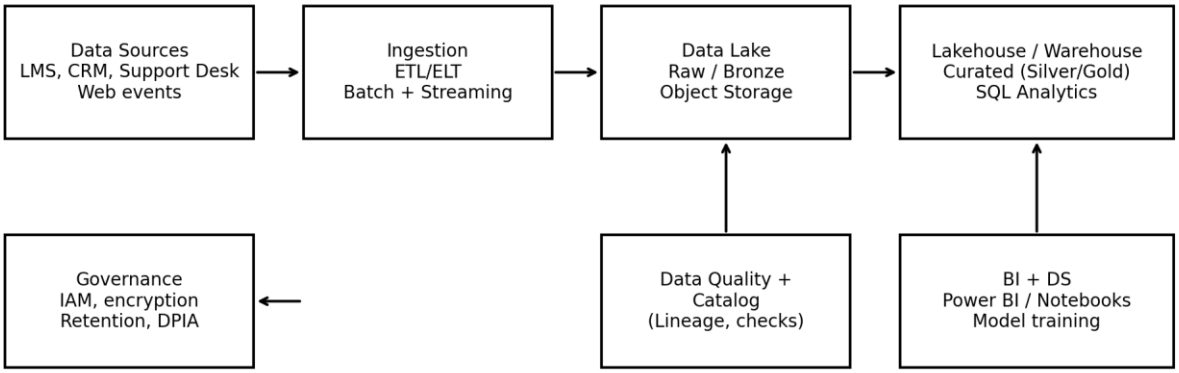
The proposed system collects three broad data types:
1) Operational support data: ticket metadata (category, timestamps, severity, SLA fields), enabling service performance management.
2) Platform telemetry: events for self-service journeys (knowledge base searches, clicks, error codes), enabling product improvement and automation evaluation.
3) Content and knowledge data: versioned policy/FAQ content and change history, enabling traceability and impact analysis when guidance changes.

### 3.2 Proposed architecture and justification of technologies

I propose a lakehouse-style architecture because it supports both traditional reporting (structured ticket tables) and big data telemetry (high-volume event logs). A lakehouse

provides scalable object storage for raw data with governance, while enabling SQL analytics and downstream machine learning. The diagram below is technology-neutral but maps cleanly to major cloud platforms.

Figure 4 illustrates the proposed data infrastructure including ingestion, storage zones, governance, and analytics layers.

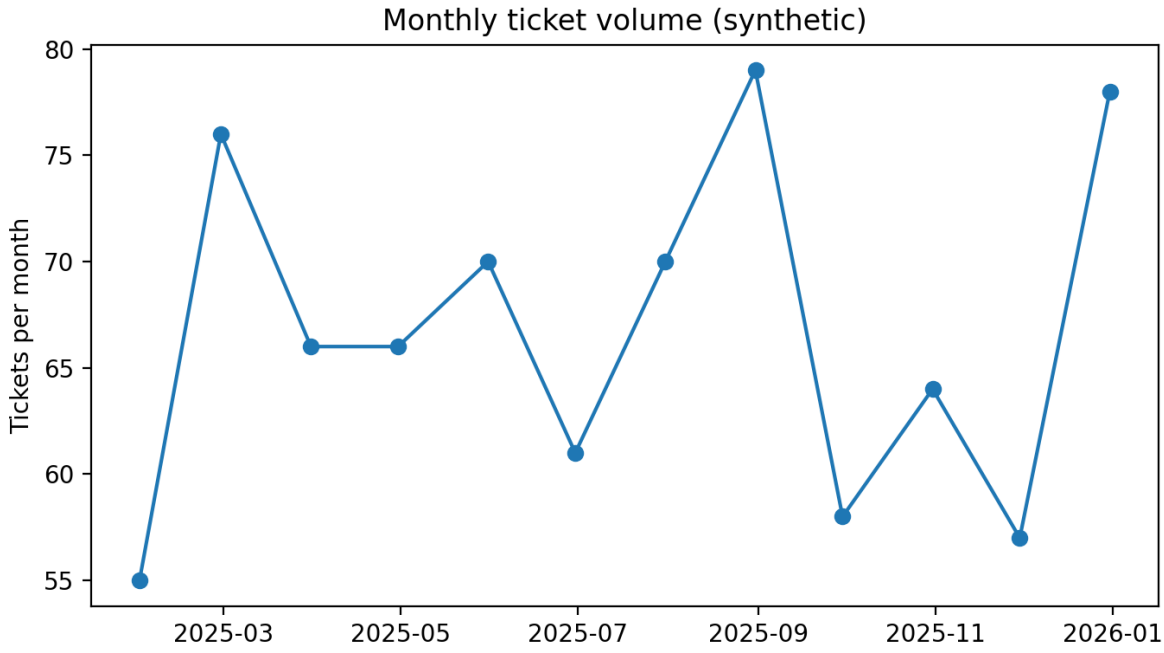



Key design choices:
- Ingestion: batch extracts from ticketing tools plus streaming telemetry for platform events.
- Storage zones: raw, curated, and semantic layers with clear data contracts.
- Governance: catalog, lineage, and access controls to support audit and quality.
- Analytics: a BI layer for dashboards and a notebook environment for advanced analysis and ML experimentation.

### 3.3 How data is stored and harnessed for analysis

Storage and analysis flow. Raw data is stored in immutable form to preserve traceability. Curated datasets apply standard cleaning and business rules (e.g., category taxonomy, SLA definitions). A semantic layer provides analyst-friendly tables for BI tools. For analysis, dashboards query the semantic layer, while data scientists use the curated zone for exploratory analysis and hypothesis testing. If machine learning is introduced later (e.g., ticket triage or satisfaction prediction), training data is drawn from curated datasets with documented provenance and bias checks.

Figure 5 provides an example of monitoring-oriented reporting: monthly ticket volume over time.




Monitoring and performance. Operational dashboards can track data freshness (last ingest time), pipeline failures, volume anomalies, and drift in category distribution. These controls support reliable decision-making and reduce the risk of acting on stale or incomplete data.

## 4. Task 4 - Legal and ethical considerations

### 4.1 Legal and ethical aspects of data in general

Legal and ethical handling of data is grounded in the UK GDPR and the Data Protection Act 2018. Key principles include: lawfulness, fairness and transparency; purpose limitation; data minimisation; accuracy; storage limitation; integrity and confidentiality; and accountability. Ethically, the organisation should avoid collecting data that is not necessary, ensure learners understand how their data is used, and design processes that respect user autonomy and avoid harm.

### 4.2 Legal and ethical aspects of the proposed system

For the proposed support analytics system, personal data may be present in user identifiers and potentially in free-text ticket fields. Controls include: pseudonymisation, access restriction, encryption, retention limits, and role-based access. Where possible, analytics should use aggregated metrics rather than identifiable records. A DPIA (Data Protection Impact Assessment) would be appropriate because the system processes behavioural telemetry and could inform decisions affecting learners.

### 4.3 Future machine learning concerns and mitigations

If machine learning is introduced (e.g., automated categorisation, prioritisation, or satisfaction prediction), additional risks emerge: bias and unfair treatment, lack of transparency, feedback loops, and automation bias where staff over-trust model outputs. Mitigations include: clear model purpose and limits, representative training data, bias testing across learner groups (where lawful and appropriate), human-in-the-loop review for high-impact decisions, and ongoing monitoring for drift and performance degradation. Model documentation (data sheets, model cards) and an audit trail support accountability.

## 5. KSB reflection on progress

The following table provides my reflection on progress against the Knowledge, Skills and Behaviours (KSBs) listed in the module brief.

| KSB | Reflection on progress |
|---|---|
| K2 | I designed a lakehouse architecture and explained how modern storage and processing choices enable scalable analytics now and future ML later, balancing cost, governance and impact. |
| K5 | I followed a structured analysis workflow: define business question, build dataset, run EDA, produce dashboard visuals, and translate insights into operational recommendations. |
| K15 | I applied engineering discipline by defining a clear artefact scope, documenting assumptions, and structuring outputs (figures, tables, repeatable steps) that could be recreated in a workplace toolchain. |
| K20 | I demonstrated collection and analysis logic for a realistic ticketing domain and produced multiple visualisations suitable for a BI dashboard. |
| K22 | I connected statistical ideas (sampling, variance, hypothesis testing, effect size) to business interpretation and explained why the chosen test was appropriate. |

| | |
|---|---|
| K24 | I identified sources of error/bias (measurement, selection, categorisation, confounding) and proposed practical controls and monitoring to minimise them. |
| K26 | I applied the scientific method by defining hypotheses, selecting a statistical test, running it, and interpreting results for strategy. |
| K27 | I proposed an instrumentation approach for collecting platform events and ticket data with quality controls and governance. |
| S1 | I used applied research logic to justify a secure, stable, scalable storage architecture with clear layers and governance mechanisms. |
| S9 | I manipulated and summarised the dataset to create decision-ready visuals and comparisons across categories and channels. |
| S10 | I selected a dataset schema that maps to real organisational processes and used methods aligned to the business questions. |
| S13 | I identified appropriate tooling choices (BI, SQL, notebooks, orchestration) and explained how they fit the computational problem. |
| S17 | I described curation and quality controls including schema validation, outlier checks, and monitoring for missingness and drift. |
| S18 | I included an infrastructure diagram and described monitoring dashboards for data freshness, pipeline health, and performance. |
| S21 | I quantified uncertainty using a 95% confidence interval and effect size, and |

| | |
|---|---|
| | discussed limitations and outliers. |
| S22 | I combined EDA and hypothesis testing to support a concrete business recommendation on self-service investment and triage. |
| S26 | I selected appropriate DS techniques (dashboard analytics and statistical inference) without over-complicating beyond the business need. |
| B3 | I demonstrated integrity by using synthetic data for GDPR reasons and by proposing privacy, security and retention controls for a real deployment. |
| B7 | I would share outputs as a reusable dashboard template, a data quality checklist, and a short internal briefing to promote consistent analytics practices. |

## 6. References (Harvard)


- ICO (n.d.) Guide to the UK GDPR. Information Commissioner's Office.
- UK Government (2018) Data Protection Act 2018.
- NIST (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology.
- W3C (2018) Web Content Accessibility Guidelines (WCAG) 2.1. World Wide Web Consortium.


## Appendix A - Dataset schema (synthetic)

| Field | Description |
|---|---|
| ticket_id | Unique identifier (synthetic) |
| month | Month of interaction |
| category | Support category |
| channel | Self-service or Human agent |
| severity | Low/Medium/High |
| resolution_time_min | Time to resolution in minutes |

| | |
|---|---|
| satisfaction | Satisfied/Unsatisfied |

## Appendix B - Evidence checklist

The module brief requests practical evidence via screenshots. In a workplace submission I would include screenshots of: (1) SQL extract queries or API calls, (2) data preparation steps, (3) Power BI dashboard pages, and (4) statistical test output. In this report I include representative dashboard figures and document the analysis steps to demonstrate the same competencies.